Bioconductor
OPEN SOURCE SOFTWARE FOR BIOINFORMATICS

Bioconductor for data science

An introduction to today's workshop

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Today’s programme

11:00 to 11:15	**Introduction to Bioconductor** Maria Doyle
11:15 to 13:00	**Wastewater metagenomics** Nick Cooley
13:00 to 14:00	**Lunch**
14:00 to 15:45	**Single-cell and complex data** Michael Lynch
15:45 to 16:00	**Discussion and wrap-up** Everyone

Taking part today

Questions are welcome

In the room: raise a hand. Online: raise a hand or use Teams chat.

Shared notes and questions

One place for everyone’s questions, useful links and troubleshooting.

A welcoming space

Respect different backgrounds and levels of experience.

Bioconductor Code of Conduct

Concerns? Speak to Maria.

What is Bioconductor?

Open-source software and a community for analysing biological data in R

Software packages	Methods for biological data analysis
Shared infrastructure	Data structures that connect tools
People and learning	Documentation, training and support

25 years of Bioconductor

25 years

Started in 2001

2,400+

Software packages

A global open-source community

Researchers, developers and educators building and sharing tools

Package scale based on the Bioconductor 3.23 release, April 2026

Your workshop team



Maria Doyle

Bioconductor Community Manager

Introduction to Bioconductor



Nick Cooley

Wastewater analysis



Michael Lynch

Single-cell and complex data analysis

Why use Bioconductor?

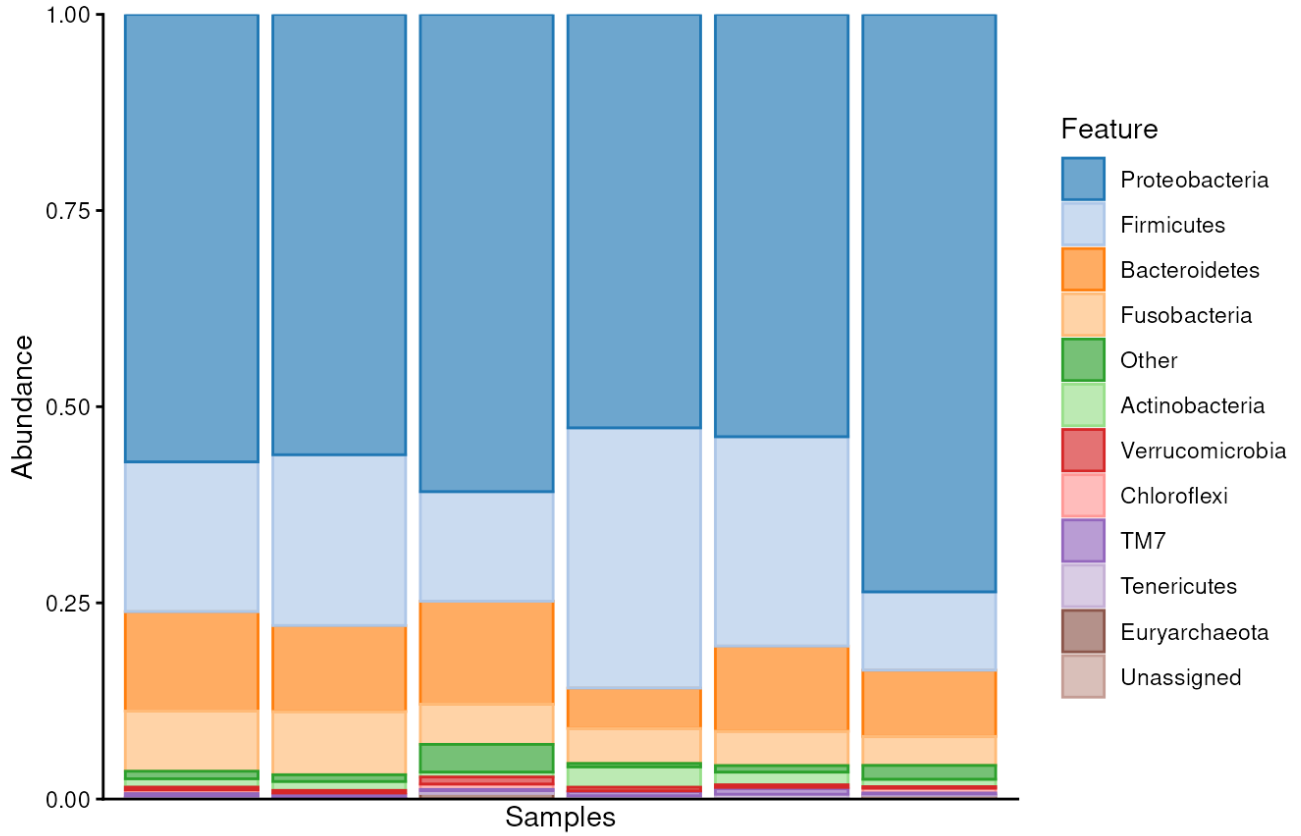
Data and context	Measurements and their descriptions stay together.
Connected tools	Packages can build on the same objects.
Documented workflows	Worked examples explain the steps.
Review and regular checks	Package review and build checks support quality.

Wastewater metagenomics

Nick Cooley

Which microbes are present?

What functions might their genes support?



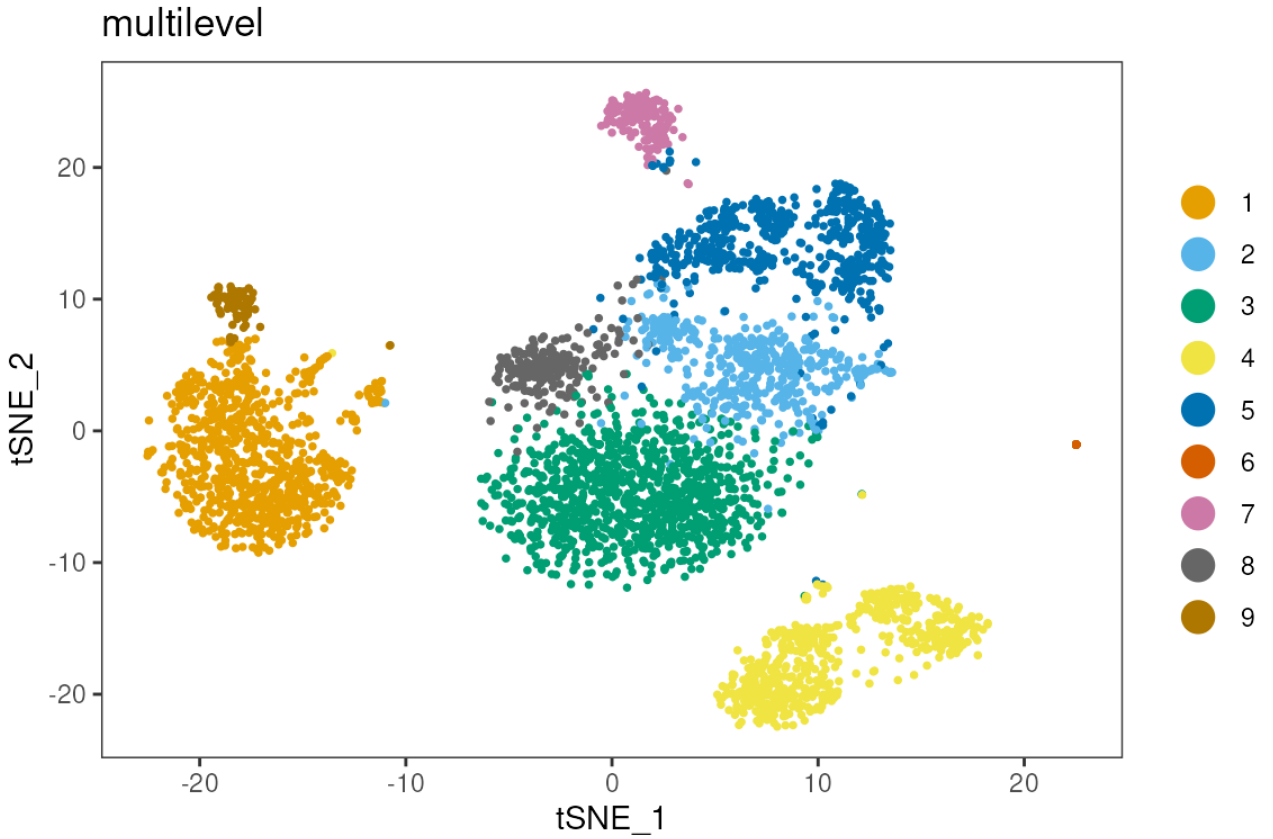
Each bar is a sample. Colours show microbial groups.

Single-cell data

Michael Lynch

How do individual cells differ?

Patterns in gene expression can help us explore cell types and states.



Each dot is a cell. Colours show clusters.

Finding a package and its guide

SingleCellExperiment

bioconductor.org/packages/SingleCellExperiment

Description	What the package does
Installation	Instructions for your R environment
Documentation	Vignettes: worked guides with code

Learning and joining the community

Workshop materials

Today's workshop page

Training

training.bioconductor.org

Questions and discussion

support.bioconductor.org

chat.bioconductor.org (Zulip)

Your questions and feedback are useful contributions.